

# DeepSeek-V3 Post-Training

Deep Dive into DeepSeek — Chapter 7

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DeepSeek-V3's post-training record is an [evidence ledger](#), not a training log.

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Goal: follow one [evidence ledger](#) from the public V3 state sequence, through packed SFT, typed rewards, R1-series target transfer, and reported outcomes, to the boundary of what remains unmeasured.

The evidence ledger, not a training log

Checkpoint vs. rendering: a matched contrast

Packed SFT: two masks, not one

Sparse customization: ESFT ledger

Reward surfaces and R1-series target transfer

Reported outcomes and the closed studies

**The evidence ledger, not a  
training log**

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## A tuple per stage, not a reconstructed trainer

$$\mathcal{S}_k = (\theta_k, I_k, F_k, U_k)$$

- $\theta_k$  named input/output checkpoint
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R1-series output enters the target-provenance ledger, not a third optimizer arrow — the sources never place it as a separately ordered stage after GRPO.

[REPORTED: DeepSeek-V3-Base: 671B total, 37B activated per token, after 14.8T-token pretraining]

## Checkpoint vs. rendering: a matched contrast

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## Four coordinates, one intended variable

$$\Delta_{\text{PT}}(x; T, d, Q) = Q(Y(\theta_{\text{post}}, T, d; x)) - Q(Y(\theta_{\text{base}}, T, d; x))$$

- Checkpoint  $\theta$ , serialization  $T$ , decoding record  $d$ , and scorer  $Q$  jointly determine a response.

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Invariant: an unmatched base/chat or local/hosted comparison is descriptive evidence, not an identified training-stage effect.

# From concept to code: rendering is part of the request

```
from DeepSeek-V3/inference/generate.py

messages.append({"role": "user", "content": prompt})
prompt_tokens = tokenizer.apply_chat_template(
    messages, add_generation_prompt=True)
...
next_token = sample(logits, temperature)
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The chat template and the generation cue are part of  $T$ ; the temperature is part of  $d$ . Neither is the checkpoint. Swap in a raw prefix and you have changed  $T$ , not just  $\theta$ . [INSPECTED:

DeepSeek-V3 inference/generate.py]



**Packed SFT: two masks, not one**

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## Sharing a sequence without sharing context

Pack sample  $A = (a_1, a_2, a_3)$  and  $B = (b_1, b_2)$  into  $z = (a_1, a_2, a_3, b_1, b_2)$ . Mask row  $t$ : which earlier position  $j$  may token  $t$  attend to?

|   |           |           |           |           |
|---|-----------|-----------|-----------|-----------|
| 0 | $-\infty$ | $-\infty$ | $-\infty$ | $-\infty$ |
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| $-\infty$ | $-\infty$ | $-\infty$ | 0         | $-\infty$ |
| $-\infty$ | $-\infty$ | $-\infty$ | 0         | 0         |

The attention mask blocks cross-sample edges; a separate loss mask  $\ell = (0, 1, 1, 0, 1)$  decides which next-token predictions own the loss. A causal-only mask would let  $b_1$  read all of  $A$  — six blocked edges here.

# From concept to code: the deterministic checkpoint

```
from code/chapter_deepseek_v3_post_training/packed_sft_printed_checkpoint.py

assert report["loss_mask"] == (0, 1, 1, 0, 1)
assert report["supervised_token_count"] == 3
assert abs(report["masked_mean_nll"] - 0.5) < 1e-12
assert report["prevented_causal_leak_edges"] == 6
```



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Small numbers, checkable by hand: three supervised tokens, mean NLL 0.5, six edges the sample-isolated mask must block. [OBSERVED: chapter-local deterministic checkpoint]

## **Sparse customization: ESFT ledger**

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## Router top- $k$ and trainable-parameter selection are different knobs

| Policy            | Trainable | Task checkpoint | Loaded |
|-------------------|-----------|-----------------|--------|
| Full SFT          | 73        | 73              | 73     |
| LoRA / QLoRA      | 8         | 8               | 81     |
| ESFT, routed only | 14        | 14              | 73     |

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Worked trace:  $\tau = 0.60$  selects seven of twelve routed experts across three toy MoE layers. Selecting them changes gradient ownership; top-2 inference still performs six routed-expert applications per token.

## Reward surfaces and R1-series target transfer

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## A typed record keeps $\perp$ separate from zero

$$\rho_i = (x, y_i, r_i^{\text{rule}}, r_i^{\text{model}}, v_i) \quad R_i = \mathcal{A}_{\text{private}}(r_i^{\text{rule}}, r_i^{\text{model}}, v_i)$$

- Rule and model rewards are reported feedback classes; the aggregation  $\mathcal{A}_{\text{private}}$  that turns them into one scalar is not.

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For rewards (1.0, 0.6, 0.4, 0.0): mean 0.5, variance 0.13, advantages  
 $\approx (1.387, 0.277, -0.277, -1.387)$  at  $\epsilon = 0$  — a tensor handoff, not a training-log row.

## Four owners a distillation claim must not collapse

$$y^{\star} = \mathcal{S}_{\text{private}}\left(x, \mathcal{Y}^{(T)}(x), c\right)$$

Teacher generates candidate responses  $\mathcal{Y}^{(T)}(x)$

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The teacher need not be present at learner inference time. A selected teacher response is supervised data — not an online reward, not proof the learner matches teacher capability.

## **Reported outcomes and the closed studies**

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## A benchmark row is a coordinate tuple before it is a score

| Artifact | Benchmark/protocol       | Score |                       |
|----------|--------------------------|-------|-----------------------|
| V3 Base  | MMLU, 5-shot accuracy    | 87.1  | [REPORTED: V3 report] |
| V3 Base  | MATH, 4-shot exact match | 61.6  | [REPORTED: V3 report] |



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| V3 Chat  | MATH-500, exact match    | 90.2  | [REPORTED: V3 report] |
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90.2 — 61.6 is not a post-training gain: different dataset, different prompting, different artifact.  
Stage attribution needs a matched checkpoint ablation the public record does not supply.

# What is closed, and why

671GB (official V3 weights, one byte/parameter) > 564GB (four H200s, aggregate)

- Base-vs-chat rendering, tool-schema adherence, sampling sensitivity, refusal classification, and judge agreement are all declared [closed-unavailable](#) under the verified device-only plan.

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No substitute-model diagnostic supplies a V3 capability, safety, sampling, or judge-agreement result. [OPEN: no local V3 behavioral study]

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Reconstruct disclosed mechanics exactly. Name every unknown owner. Never let a substitute artifact answer a V3 question.

Next: the R1-Zero chapter turns pure RL and verifiable rewards into a fully derivable objective — because that model's public sources support it.